refine

Below is a professionally structured and refined version of our ERP requirements. I have organized the modules, improved the terminology, clarified the responsibilities, and connected the workflows between departments.

ERP System – Main Modules & Functional Requirements

Medium-sized, modular ERP system for a Gt printing press and printing manufacturing company

1. Finance Management Module

The Finance Management Module manages all company financial activities, expenses, revenue, profitability, assets, liabilities, and financial reporting. It should be integrated with HR, Store, Purchasing, Operations, and CRM/Orders.

1.1 Financial Dashboard

- Total company income and expenditure.
- Total revenue, expenses, net profit, and net loss.
- Profit margin and production profitability.
- Order-based and project-based financial performance.
- Production costs, material costs, and wastage costs.
- Transportation and delivery expenses.
- Outstanding payments and receivables.
- Payable amounts and company liabilities.
- Financial trends and performance charts.

1.2 Income and Expense Management

- Record and manage all company income and expenses.
- Support fixed and variable expenses.

- Support manual and dynamic cost allocation.
- Configure income and expense categories.
- Allocate costs to individual orders, projects, departments, and production activities.
- Track direct and indirect costs.
- Manage transportation, operational, and miscellaneous expenses.
- Track production wastage and financial losses.

1.3 Cost and Profitability Management

- Calculate the actual cost of each order.
- Calculate raw material and production costs.
- Calculate labor, machine operation, and transportation costs.
- Compare estimated costs against actual costs.
- Calculate gross profit, net profit, and profit margins.
- Configure manual or dynamic profit margin percentages.
- Analyze order profitability.
- Calculate ROI (Return on Investment).

1.4 Asset Management

- Fixed asset management.
- Consumable asset management.
- Asset registration and categorization.
- Asset assignment and ownership tracking.
- Asset depreciation and valuation.
- Asset maintenance and replacement tracking.

1.5 Tax Management

- Tax payable management.
- Tax receivable management.
- Tax calculation and tracking.
- Tax-related financial reports.
- Tax transaction history.

1.6 Payroll and Financial Integration

- Employee salary and payroll processing.
- Overtime payments and deductions.
- Bonuses, allowances, and penalties.
- Payroll approval and payment tracking.
- Integration with the HR module.
- Automatic recording of payroll expenses in Finance.

2. Human Resource Management (HR) Module

The HR Module manages employees, recruitment, attendance, leave, payroll coordination, administrative activities, and overall workforce management.

2.1 HR Dashboard

- Total employees.
- Active and inactive employees.
- Attendance overview.
- Present, absent, and late employees.
- Pending leave requests.
- Overtime overview.
- Payroll summary.
- Recruitment and internship statistics.
- Department and employee performance indicators.

2.2 Employee Management

- Employee registration and profile management.
- Employee directory.
- Unique employee ID generation.
- Personal and professional information.
- Employee documents and contracts.
- Department and position assignment.
- Employment status management.
- Employee history and performance records.

2.3 Attendance Management

- Daily attendance tracking.
- Check-in and check-out records.
- Late arrival and early departure tracking.
- Absence management.
- Attendance correction requests.
- Attendance reports and analytics.
- Integration with payroll calculations.

2.4 Leave Management

- Employee leave requests.
- Different leave categories.
- Leave balance management.
- Multi-level approval workflow.
- Leave history and reports.
- Real-time leave request notifications.

2.5 Payroll Integration

- Employee salary information.
- Salary structure management.
- Overtime calculations.
- Allowances and deductions.
- Payroll preparation and approval.
- Direct integration with Finance.
- Salary payment history.

2.6 Internship Management

- Internship application management.
- Internship acceptance and registration.
- Intern profile and document management.
- Department and supervisor assignment.
- Internship duration tracking.
- Attendance and performance tracking.
- Internship completion and certification records.

2.7 Employee Clearance Management

- Employee resignation and termination clearance.
- Department-based clearance processes.
- Store property return verification.
- Financial clearance.
- HR final approval.
- Clearance history and documentation.

2.8 Recruitment and Job Management

- Internal and external job posting.
- Job vacancy management.
- Online job application portal.
- Applicant profile management.
- CV and document uploads.
- Applicant screening and shortlisting.
- Interview scheduling and evaluation.
- Recruitment status tracking.

2.9 Overtime Management

- Overtime request submission.
- Overtime approval workflow.
- Actual overtime hour tracking.
- Overtime payment calculation.
- Department-based overtime reports.
- Integration with payroll.

2.10 Communication and Administrative Tools

- Real-time notifications.
- Internal messaging and communication.
- Employee and department announcements.
- Notes and reminders.
- HR settings and configurations.
- Advanced search, filtering, and reporting.

2.11 AI-Powered HR Assistant

- Answer HR-related questions.
- Retrieve employee information based on permissions.
- Generate HR reports and summaries.
- Analyze attendance and overtime patterns.
- Assist with recruitment and applicant screening.
- Identify unusual attendance patterns.
- Provide HR management insights.

2.12 AI Administrative Assistant

- Generate administrative reports.
- Summarize company activities.
- Assist with company planning.
- Track pending administrative tasks.
- Identify operational bottlenecks.
- Provide decision-support insights based on company data.

3. Store and Inventory Management Module

The Store Module manages all company inventory, raw materials, consumables, fixed assets, stock movements, and internal material requests.

It must be fully integrated with Purchasing, Finance, Operations, and Administration.

3.1 Store Dashboard

- Total stock items.
- Total inventory value.
- Available and reserved stock.
- Low-stock and out-of-stock items.
- Incoming and outgoing stock.
- Pending material requests.
- Pending stock approvals.
- Stock movement overview.

- Inventory valuation reports.

3.2 Inventory Management

- Product and material registration.
- Stock categorization.
- Fixed and consumable inventory classification.
- Raw material management.
- Finished goods and other inventory management.
- Multiple warehouse and storage location support.
- Stock quantity and valuation tracking.
- Batch and unit management.
- Minimum and maximum stock level configuration.

3.3 Real-Time Stock Monitoring

- Real-time stock quantity updates.
- Automatic low-stock alerts.
- Out-of-stock notifications.
- Stock expiry alerts where applicable.
- Stock movement notifications.
- Automatic purchasing requests for low-stock items.
- Notify Purchasing and Administration when stock reaches its reorder level.

3.4 Stock Inward Management

- Goods receiving registration.
- Goods Received Voucher (GRV) integration.
- Supplier and purchase order references.
- Quantity and quality verification.
- Receiving documentation.
- Storekeeper confirmation.
- Administrative approval where required.
- Automatic inventory updates.

3.5 Stock Outward Management

- Raw material requests from production.

- Employee and department material requests.
- Support staff property requests.
- staff proporties return with approval
- Stock issue and release management.
- Request approval workflow.
- Quantity verification.
- Stock release documentation.
- Automatic stock deduction.
- Complete inward and outward transaction history.

3.6 Stock Request and Approval Workflow

- Employees submit material requests.
- Department managers review requests.
- Administration approves or rejects requests.
- Store verifies availability.
- Approved requests are issued.
- If stock is insufficient, a purchasing request is generated.
- Purchasing receives a real-time notification.

3.7 Finance and Purchasing Integration

- Track inventory purchase costs.
- Track total inventory valuation.
- Connect received goods with purchase transactions.
- Record inventory expenses in Finance.
- Track supplier-related outstanding payments.
- Provide purchasing cost reports.

3.8 Communication and Reporting

- Internal messaging.
- Real-time notifications.
- Notes and reminders.
- Store settings.
- Stock movement reports.
- Inventory valuation reports.

- Advanced search and filtering.
- AI-powered inventory assistant.

3.9 AI Inventory Assistant

- Identify frequently consumed materials.
- Analyze historical stock movements.
- Predict potential stock shortages.
- Recommend reorder quantities.
- Identify unusual stock consumption.
- Generate inventory reports and summaries.

4. Purchasing and Procurement Management Module

The Purchasing Module manages purchasing requests, suppliers, procurement processes, quotations, public auctions, purchase orders, goods receiving, and procurement expenses.

It must be integrated with Store, Finance, and Administration.

4.1 Purchasing Dashboard

- Total purchases.
- Monthly purchasing expenses.
- Pending purchasing requests.
- Approved and rejected requests.
- Pending purchase orders.
- Supplier payment status.
- Credit purchase overview.
- Transportation expenses.
- Purchasing performance reports.

4.2 Purchase Request Management

- Receive purchasing requests from Store.
- Receive requests from other authorized departments.

- Create manual purchasing requests.
- Review stock shortage notifications.
- Submit purchase requests for administrative approval.
- Approve or reject purchasing requests.
- Track request progress.
- Convert approved requests into purchase orders.

4.3 Supplier Management

- Supplier registration.
- Supplier profiles and contact information.
- Supplier categorization.
- Supplier transaction history.
- Supplier quotation management.
- Supplier performance evaluation.
- Supplier payment and credit tracking.

4.4 Purchase Order Management

- Create purchase orders.
- Select suppliers and products.
- Record quantity and unit prices.
- Calculate total purchasing costs.
- Manage discounts and additional charges.
- Administrative approval.
- Purchase order tracking.
- Purchase history.

4.5 Goods Received Voucher (GRV)

- Generate GRV for purchased items.
- Verify received quantities.
- Record damaged or missing items.
- Compare received goods with purchase orders.
- Record receiving dates and supplier details.
- Generate GRV documents.
- Connect GRV with Store inventory.

- Support partial goods receiving.

4.6 Credit and Transportation Cost Management

- Manage credit-based purchases.
- Record supplier debts.
- Track payment deadlines.
- Manage partial and full payments.
- Track transportation and delivery expenses.
- Allocate transportation costs to purchased products.
- Calculate actual landed purchasing costs.
- Synchronize financial transactions with Finance.

4.7 Public Auction Management

- Register public auction opportunities.
- Manage auction items and specifications.
- Record auction prices and conditions.
- Compare prices from different suppliers or auction opportunities.
- Track auction deadlines.
- Record bidding and purchasing decisions.
- Generate auction comparison reports.
- Approval before final purchase commitment.

4.8 Reporting and Communication

- Purchasing reports.
- Supplier reports.
- Purchase cost analysis.
- Credit and outstanding payment reports.
- Transportation cost reports.
- Notes and reminders.
- Internal messaging.
- Real-time notifications.
- Advanced search and filtering.
- AI-powered purchasing assistant.

4.9 AI Purchasing Assistant

- Analyze historical purchasing prices.
- Compare supplier pricing.
- Identify cost-saving opportunities.
- Recommend purchasing quantities.
- Analyze supplier performance.
- Identify unusual purchasing expenses.
- Generate procurement summaries.

5. Machine and Production Operations Module

The Machine and Production Operations Module manages manufacturing activities, machine operations, raw material consumption, maintenance, production wastage, and production performance.

5.1 Machine Management

- Machine registration.
- Machine identification and categorization.
- Machine specifications.
- Machine status tracking.
- Machine assignment to production jobs.
- Machine operating history.
- Machine utilization reports.

5.2 Maintenance Management

- Preventive maintenance scheduling.
- Daily machine maintenance checklists.
- Maintenance request submission.
- Breakdown and repair registration.
- Maintenance approval and tracking.
- Maintenance cost recording.
- Maintenance history.
- Machine downtime reports.

5.3 Raw Material Request Management

- Production-based raw material requests.
- Request materials based on approved orders.
- Check available inventory.
- Submit material requests to Store.
- Administrative approval where required.
- Track issued quantities.
- Record actual material consumption.
- Return unused materials to Store.

5.4 Production Wastage Management

- Register production wastage.
- Record wasted raw materials.
- Record damaged or defective products.
- Categorize wastage reasons.
- Record responsible production stages.
- Calculate wastage quantities and costs.
- Submit wastage records for approval.
- Generate wastage reports.
- Integrate wastage costs with Finance.

5.5 Production Monitoring

- Production job assignment.
- Production stage tracking.
- Machine operator assignment.
- Production start and completion times.
- Planned versus actual production.
- Production quantity tracking.
- Production efficiency reports.

6. CRM, Order and Sales Management Module

The CRM, Order and Sales Module manages the complete customer journey, from lead generation and customer follow-up to quotations, order processing,

production, invoicing, and payment collection.

6.1 CRM Dashboard

- Total customers.
- New leads.
- Active opportunities.
- Pending follow-ups.
- Confirmed orders.
- Completed orders.
- Outstanding invoices.
- Total sales revenue.
- Customer payment status.
- Sales performance reports.

6.2 Lead Management

- Lead registration.
- Lead source tracking.
- Customer information management.
- Lead categorization.
- Lead assignment to sales representatives.
- Lead status tracking.
- Lead conversion into customers.

6.3 Customer Management

- Customer registration.
- Customer profile and contact details.
- Individual and business customer categories.
- Customer transaction history.
- Customer order history.
- Customer credit history.
- Customer communication records.

6.4 Follow-up Management

- Customer follow-up scheduling.

- Follow-up reminders.
- Sales representative assignment.
- Communication history.
- Follow-up status tracking.
- Real-time notifications.

6.5 Service Management

- Manage printing services.
- Define service categories.
- Configure service specifications.
- Manage printing types and options.
- Configure material and finishing requirements.
- Manage service pricing.

6.6 Order Management

- Create customer orders.
- Generate unique order numbers.
- Assign customers and sales representatives.
- Record printing specifications.
- Attach designs and reference files.
- Set order quantities and deadlines.
- Track order status.
- Manage order revisions.
- Connect orders with production operations.

6.7 Dynamic Production Workflow

- Configure different workflows based on the printing service.
- Define production stages.
- Assign responsible departments and employees.
- Track production progress in real time.
- Manage approvals between production stages.
- Record stage completion dates.
- Track delays and bottlenecks.

Example workflow:

Printing Order Workflow

1. Customer Request & Lead

   Customer requirements and printing specifications.

2. Quotation & Proforma

   Estimate price, materials, and delivery time.

3. Order Confirmation

   Customer approval and initial payment.

4. Design Department

   Design preparation and customer approval.

5. Production & Machine

   Raw material allocation and machine production.

6. Finishing Department

   Cutting, binding, lamination, packaging, and quality checking.

7. Quality Control

   Final inspection and approval.

8. Delivery & Collection

   Customer delivery and payment settlement.

9. Order Completion

   Close order and calculate actual profitability.

6.8 Price Estimation and Cost Registration

- Dynamic printing price estimation.
- Manual price configuration.
- Raw material cost calculation.

- Machine operation cost calculation.
- Labor cost calculation.
- Finishing and packaging costs.
- Transportation and delivery costs.
- Direct and indirect cost allocation.
- Estimated versus actual production cost comparison.

6.9 Proforma and Invoice Management

- Generate proforma invoices.
- Generate official invoices.
- Unique invoice numbering.
- Customer and order information integration.
- Tax and discount calculations.
- Partial payment records.
- Invoice approval workflow.
- PDF document generation.
- Invoice history and tracking.

6.10 Credit and Payment Collection

- Customer credit management.
- Credit limit configuration.
- Outstanding balance tracking.
- Partial and full payment registration.
- Payment method management.
- Payment collection tracking.
- Payment reminders.
- Customer debt reports.
- Integration with Finance.

6.11 Sales and Profitability Reports

- Daily, weekly, monthly, and yearly sales.
- Customer-based sales reports.
- Order-based profitability.
- Estimated versus actual cost.

- Revenue and expense comparison.
- Sales representative performance.
- Pending and completed orders.
- Order delay analysis.
- Profit margin analysis.

6.12 AI-Powered Sales Assistant

- Generate customer quotations.
- Assist with price estimation.
- Analyze customer purchasing patterns.
- Identify profitable services.
- Summarize customer interactions.
- Recommend follow-up actions.
- Generate sales and order reports.

7. Cross-Module System Requirements

These are shared capabilities that should work consistently throughout the entire ERP system rather than being implemented independently in every module.

7.1 Centralized Administration and Approval

- Centralized administration dashboard.
- User and department management.
- Role-based access control (RBAC).
- Permission-based module access.
- Configurable approval workflows.
- Multi-level approval support.
- Audit logs for important actions.
- Transaction approval history.

7.2 Real-Time Communication

- Internal messaging.
- Department-based communication.
- Real-time notifications.
- System announcements.
- Notes and reminders.
- Notification preferences.
- Read and unread status.

7.3 Reporting and Analytics

- Centralized reporting engine.
- Advanced search and filtering.
- Date-range filtering.
- Department-based reports.
- Export to PDF and Excel.
- Custom report generation.
- Dashboard charts and statistics.
- Historical data analysis.

7.4 AI Integration

- Context-aware AI assistant for each department.
- Centralized AI administrative assistant.
- AI-powered reporting and summaries.
- Natural-language queries over authorized company data.
- Data-driven recommendations.
- Permission-aware AI access.
- Human approval for important financial and administrative actions.

7.5 System Configuration

- Company information and branding.
- Department and organizational structure.
- User and role configuration.
- Currency and tax configuration.
- Approval workflow configuration.

- Notification settings.
- Financial settings.
- Inventory and stock settings.
- Audit and activity logs.

8. Core Module Integration Architecture

The most important requirement is that all modules share connected business data. For example, purchasing, stock, production, and finance must not operate as separate systems.

ERP Module Integration

CENTRAL ADMINISTRATION

Users · Permissions · Approvals · Configuration · Audit Logs

HR

Employees · Attendance · Payroll

FINANCE

Income · Expense · Tax · ROI

STORE

Inventory · Inward · Outward

PURCHASING

Suppliers · PO · GRV

OPERATIONS

Machines · Production · Waste

CRM & SALES

Customers · Orders · Invoices

Shared data, events and business workflows

CENTRALIZED ERP DATABASE & SERVICES

Authentication · APIs · Business Logic · Notifications · Reporting · AI

Important integration examples

Event	Connected modules
Employee salary processed	HR → Finance
Overtime approved	HR → Finance
Stock reaches minimum level	Store → Purchasing → Admin
Purchase approved	Purchasing → Finance
Goods received	Purchasing → Store
Materials issued to production	Store → Operations
Production wastage registered	Operations → Store → Finance
Customer order confirmed	CRM → Operations
Order completed	Operations → CRM
Customer payment collected	CRM → Finance
Order cost calculated	Operations → Finance
Customer invoice generated	CRM → Finance

9. Recommended Development Priorities

For a medium-sized printing press ERP, I recommend building this as a modular monolithic system initially, with well-separated business modules and shared services. This keeps development, deployment, maintenance, and debugging manageable without sacrificing scalability.

Phase	Modules	Priority
Phase 1	Authentication, Admin, HR foundation	Essential
Phase 2	Store and Purchasing	Essential
Phase 3	CRM, Sales, Orders and Invoices	Essential
Phase 4	Machine and Production Operations	Essential
Phase 5	Finance and complete integrations	Essential
Phase 6	Advanced reports and analytics	High
Phase 7	AI assistants and predictive analytics	Later

Important architectural recommendation: Build the financial transaction engine, approval system, audit logs, and notification infrastructure early. Every module will depend on these shared capabilities.

The AI assistants should initially be read-only and permission-aware. They can generate reports, answer questions, and provide recommendations. Financial transactions, approvals, stock adjustments, and other sensitive actions must require authorized human confirmation.

The overall objective: Build a centralized ERP platform that controls the complete business lifecycle of a printing press company — from customer acquisition and order creation to purchasing, inventory, production, delivery, payment collection, payroll, and financial profitability analysis.